

Estimating Differential Turnout and Mobilization Gaps

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Who stays home can decide who governs

- ▶ Turnout is unequal across citizens, groups, and parties – democracy's “unresolved dilemma” (Lijphart, 1997)
- ▶ Under **first-past-the-post**, small differences in votes become large differences in seats
- ▶ Since turnout gaps are likely partisan, election outcomes reflect mobilization capacity, not only voter preferences, and these gaps might determine election outcomes

The measurement problem

- ▶ Official returns report **votes cast**, not abstainers' party preferences
- ▶ Surveys ask about turnout – but respondents **over-report** participation by 10–15pp (Bernstein, Chadha & Montjoy 2001)
- ▶ Over-reporting is correlated with education, income, and partisanship
- ▶ Constituency-level survey estimates become noisy precisely where FPTP outcomes are decided
- ▶ Ecological inference (King 1997; Katz & King 1999) helps – but is parametric and hard to generalize across time and space

The key challenge is to infer the preferences of abstainers without relying on biased turnout self-reports

This paper

- ▶ Develops a method to recover **party-specific turnout** from official election returns
- ▶ Requires only constituency-level vote totals and registered electorates
- ▶ Applies the method to UK general elections, 2010–2024
- ▶ Uses the estimates to build **universal-turnout counterfactuals**

The core idea is an accounting constraint: latent party support must sum to the electorate in each constituency

Main findings

- ▶ Conservative supporters turn out at **systematically higher** rates than Labour supporters
- ▶ UKIP/Brexit/Reform UK have large **latent** support but low estimated turnout
- ▶ Universal-turnout counterfactuals would have changed government formation in **2010, 2015, and 2017**
- ▶ Survey-based measures agree for major parties; diverge for smaller or geographically diffuse parties

Partisan turnout gaps are large enough to ultimately affect election results

Literature

1. **Unequal participation & democratic consequences**

Rosenstone & Hansen (1993); Verba, Schlozman & Brady (1995); Lijphart (1997); Franklin (2004); Leighley & Nagler (2014); Tomas-Valiente, Cinar, Foos & John (2026)

2. **Partisan effects of differential turnout**

Bernhagen & Marsh (2007); Hansford & Gomez (2010)

3. **Ecological inference**

Robinson (1950); Goodman (1953); King (1997); Katz & King (1999); Greiner & Quinn (2009)

4. **Compulsory voting**

Birch (2009); Bol & Giani (2021)

Why the United Kingdom?

- ▶ 650 single-member constituencies; first-past-the-post
- ▶ FPTP **magnifies** small vote differences into large seat differences
- ▶ Multiple nationally viable parties plus strong regional parties (SNP, Plaid Cymru, NI parties)
- ▶ 2010–2024 covers coalition politics, the Brexit cleavage, the SNP rise, UKIP/Brexit/Reform, and the 2024 Labour landslide
- ▶ About **27% of constituencies decided by margins under 10pp** – a high-leverage setting for mobilization

Data

- ▶ Official House of Commons Library returns for 2010, 2015, 2017, 2019, 2024
- ▶ For each constituency-election:
 - ▶ votes for every candidate
 - ▶ invalid ballots
 - ▶ registered electorate
- ▶ Party labels harmonized across elections
- ▶ UKIP, the Brexit Party, and Reform UK treated as one right-populist Eurosceptic family
- ▶ Constituency turnout: **40–82%**; mean $\approx 65\%$

Why the data are informative about differential turnout

- ▶ Vote shares vary across constituencies within a region
- ▶ Party-specific turnout is allowed to differ by party and region
- ▶ Each constituency provides one accounting equation
- ▶ Within a region, many constituencies jointly identify a small number of party turnout rates

Constituency variation turns official returns into an over-identified system for estimating party-specific turnout

Method setup

For constituency i , party j , region $r(i)$:

V_{ij} = valid votes for party j

E_i = registered electorate

τ_{rj} = turnout rate of supporters of party j in region r

Latent supporters:

$$S_{ij} = \frac{V_{ij}}{\tau_{r(i)j}}$$

The accounting identity

Within each constituency, supporters across all parties must add up to the electorate:

$$\sum_j S_{ij} = E_i \implies \sum_j \frac{V_{ij}}{\tau_{r(i)j}} = E_i$$

Letting $g_{rj} = 1/\tau_{rj}$:

$$\sum_j V_{ij} g_{r(i)j} = E_i$$

Once turnout is expressed inversely, the unknown party turnout rates enter the model linearly

Estimation

For each region r , stack constituency equations: $A_r g_r \approx b_r$, with

- ▶ A_r : constituency-by-party vote matrix
- ▶ b_r : vector of electorates

Solve constrained least squares:

$$\min_{g_r} \|A_r g_r - b_r\|^2 \quad \text{s.t.} \quad g_{rj} \geq 1/\tau_{\max}.$$

Recover turnout: $\hat{\tau}_{rj} = 1/\hat{g}_{rj}$

Baseline cap: $\tau_{\max} = 0.90$; Robustness: 0.85/0.95

The method reduces differential turnout estimation to constrained least squares

Identification and uncertainty

- ▶ Identification comes from variation in party vote totals across constituencies within a region – it relies on within-region variation and a common regional turnout rate by party
- ▶ System is **over-determined**: Many equations (650), few party-by-region parameters ($15 \times 12 = 180$)
- ▶ The cap $\tau_{\max}$ prevents implausibly high estimates (e.g., illness, registration, ineligibility)
- ▶ Uncertainty: nonparametric bootstrap resampling constituencies within regions, $B = 10,000$
- ▶ 95% CIs for major parties typically span only **0.03–0.05**; wider for niche/regional parties

Counterfactual exercise

- ▶ Estimated latent supporters: $\hat{S}_{ij} = V_{ij} / \hat{\tau}_{r(i)j}$
- ▶ Universal-turnout counterfactual: $\tilde{V}_{ij} = \hat{S}_{ij}$
- ▶ Apply the FPTP rule to $\tilde{V}_{ij}$ in every constituency to obtain hypothetical seat allocations
- ▶ Identify constituencies that flip to a different party under full mobilization

The counterfactual asks how seats would change if observed supporters were fully mobilized

How to interpret the counterfactuals

- ▶ Not a real forecast of compulsory voting – even compulsory voting rarely yields 100% turnout
- ▶ Holds party preferences, tactical voting, candidate entry, and campaigns fixed
- ▶ More reliable for major parties; smaller parties tend to have wider confidence intervals

These are upper-bound exercises, not forecasts of how campaigns or voters would behave under compulsory voting

Aggregate turnout estimates, 2010–2024 (cap $\tau_{\max} = 0.90$)

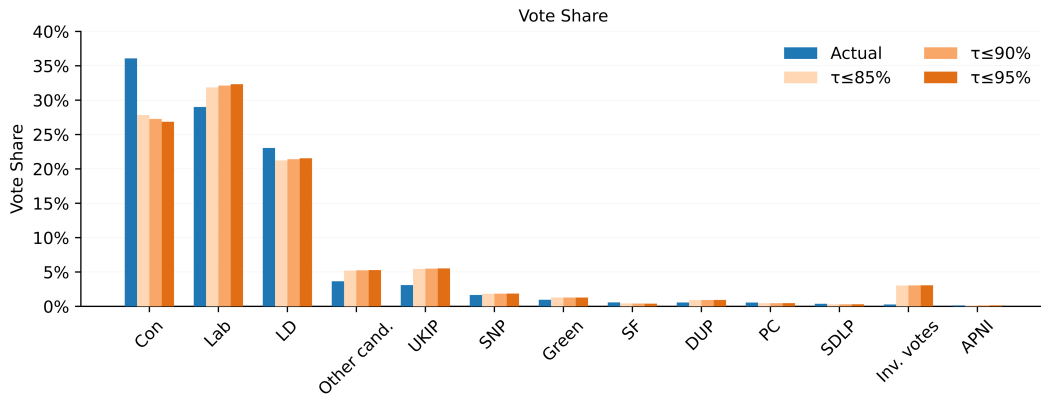
	2010	2015	2017	2019	2024
Conservative	0.85	0.87	0.80	0.67	0.90
Labour	0.58	0.66	0.69	0.66	0.68
Liberal Democrats	0.69	0.76	0.84	0.89	0.79
UKIP/Brexit/Reform UK	0.36	0.41	0.28	0.30	0.32
Green	0.48	0.58	0.73	0.74	0.59
SNP	0.57	0.62	0.47	0.64	0.49
Plaid Cymru	0.77	0.90	0.80	0.82	0.64
N. Ireland parties	0.59	0.66	0.64	0.61	0.69
Others	0.45	0.60	0.61	0.73	0.49
Invalid votes	0.06	0.06	0.04	0.08	0.03

Source: Author's calculations from official election returns; baseline cap $\tau_{\max} = 0.90$.

Aggregate turnout estimates, 2010–2024 (cap $\tau_{\max} = 0.90$)

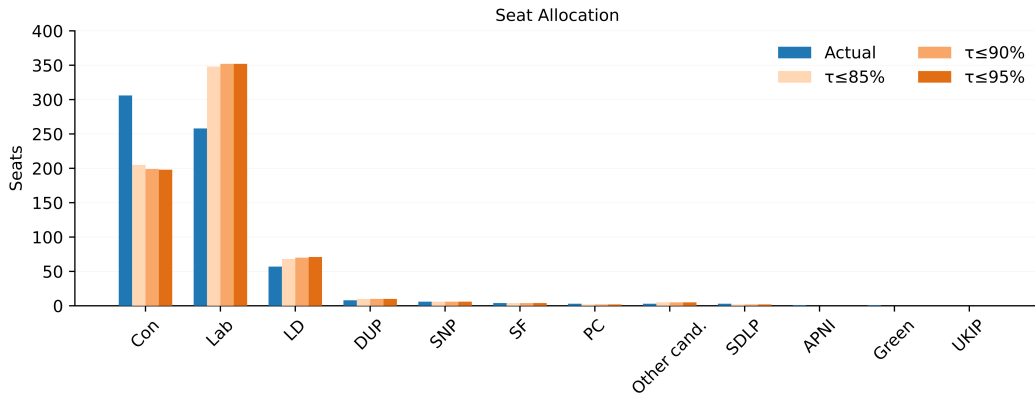
- ▶ Conservative turnout advantage is persistent: ~ 10 – 20 points over Labour in every election
- ▶ Labour turnout is stable but below Conservative even in its strongest year (2017, 69%)
- ▶ UKIP/Brexit/Reform UK have the largest mobilization deficit – never above 41%
- ▶ Liberal Democrat and Green turnout is high in some cycles, but support is electorally inefficient when diffuse

2010: turnout gaps and the coalition that followed



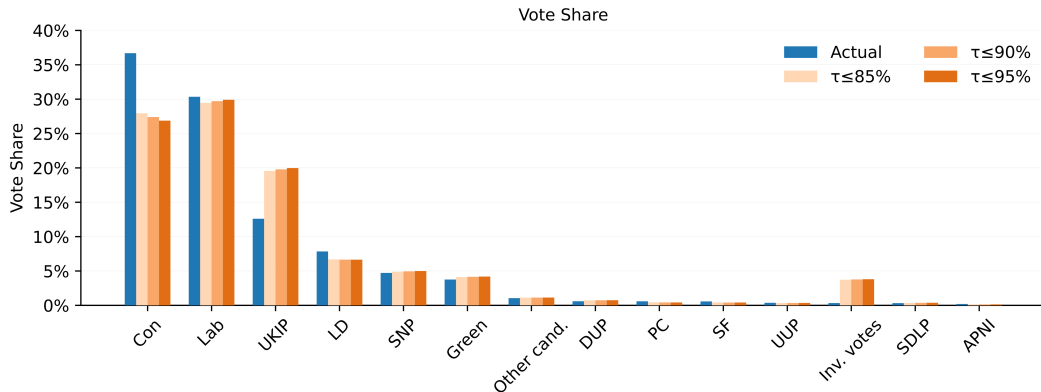
Source: Author's calculations from official election returns.

2010: turnout gaps and the coalition that followed



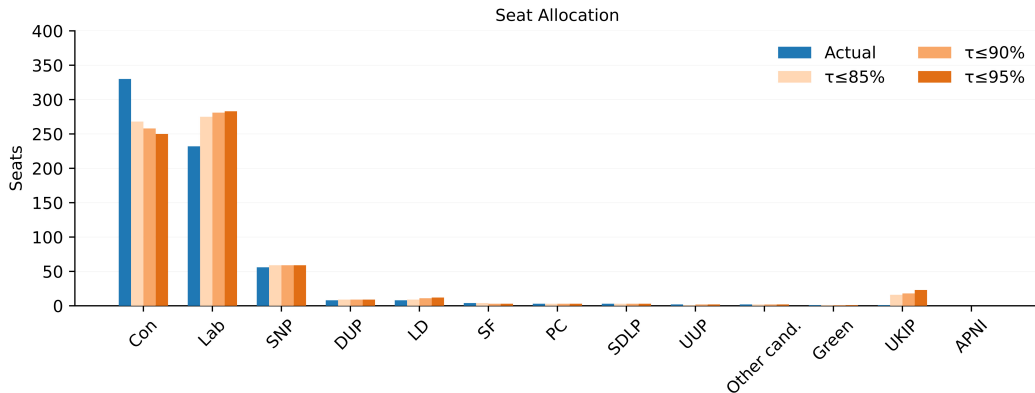
Source: Author's calculations from official election returns.

2015: the Conservative majority depended on mobilization



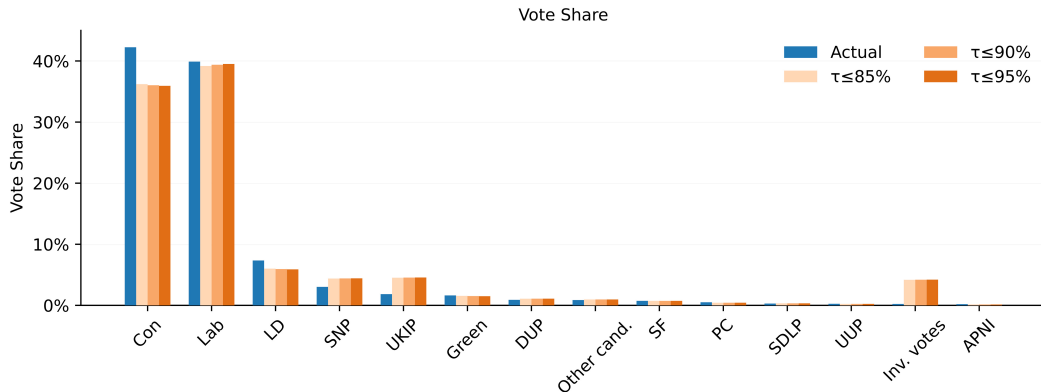
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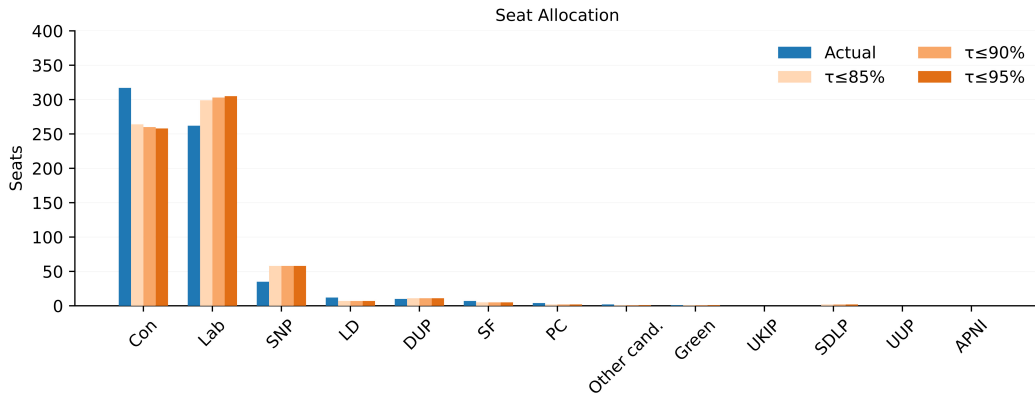
Source: Author's calculations from official election returns.

2017: the largest counterfactual reversal



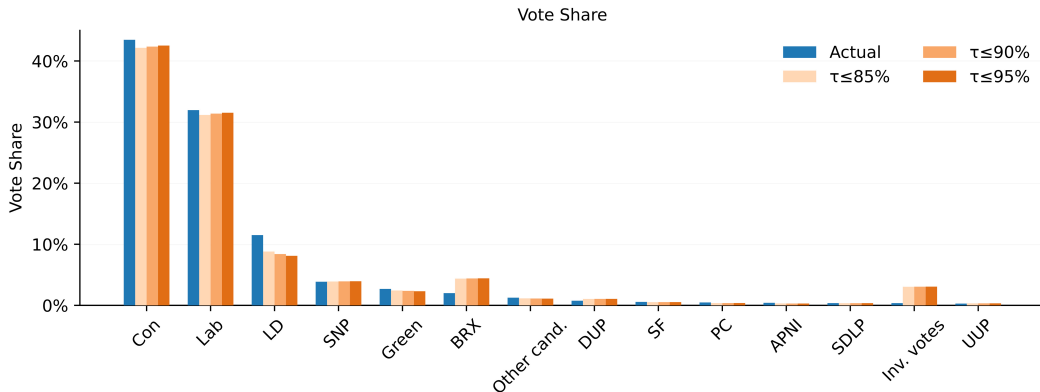
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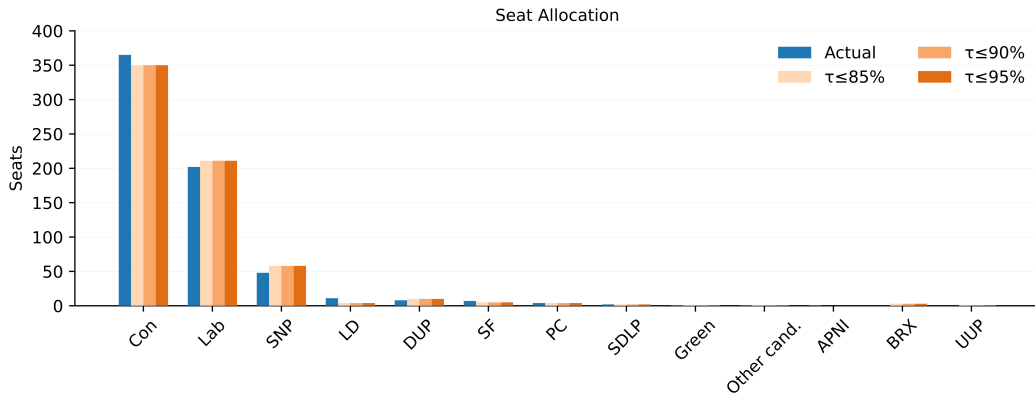
Source: Author's calculations from official election returns.

2019: outcome not reversed, but margins reshaped



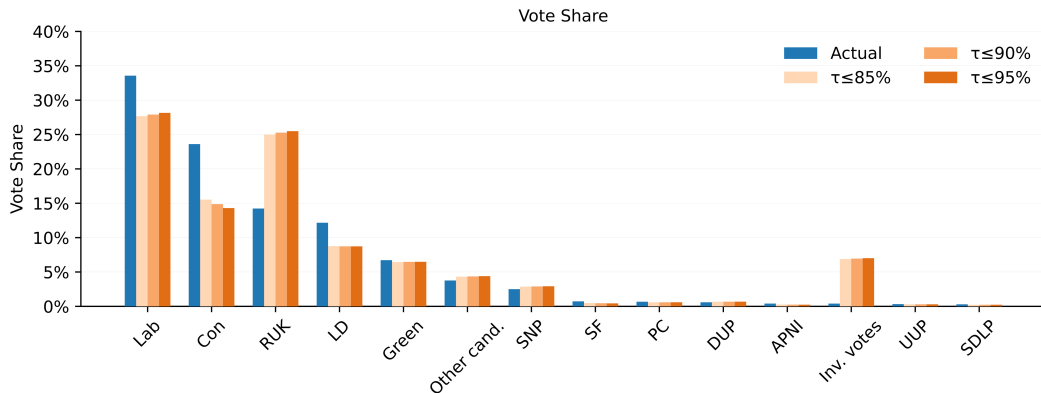
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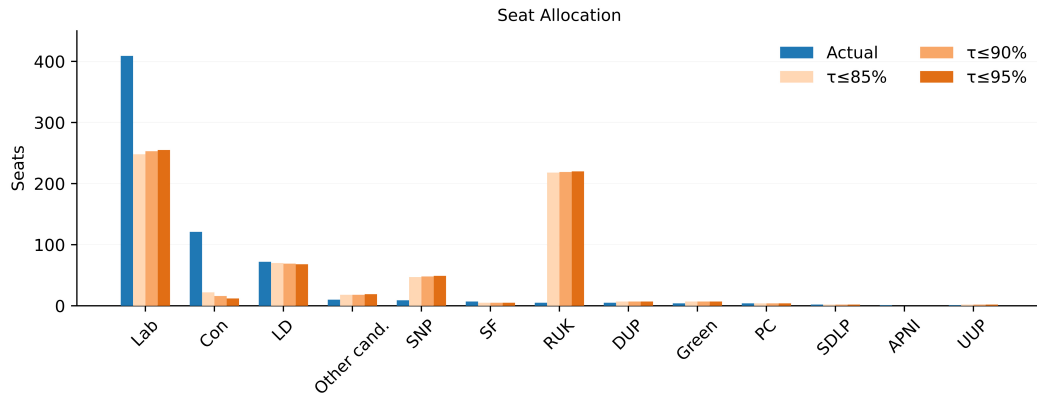
Source: Author's calculations from official election returns.

2024: Reform UK's large latent vote



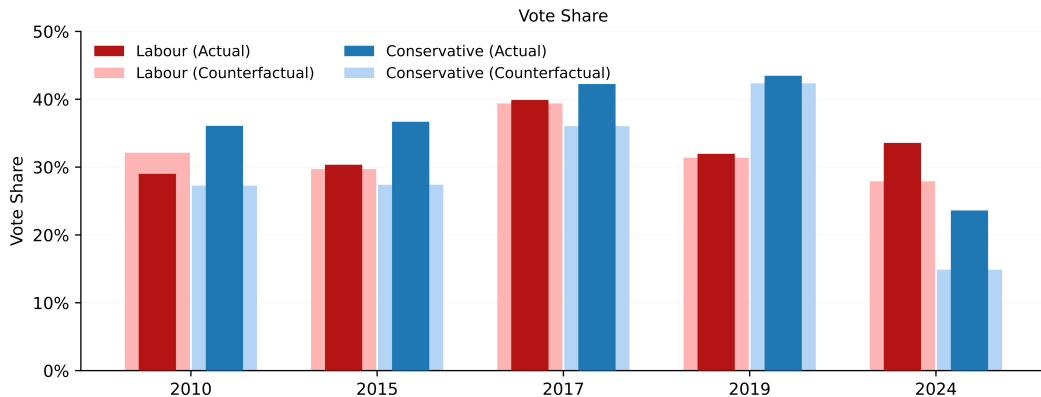
Source: Author's calculations from official election returns.

2024: Massive Reform UK's latent base



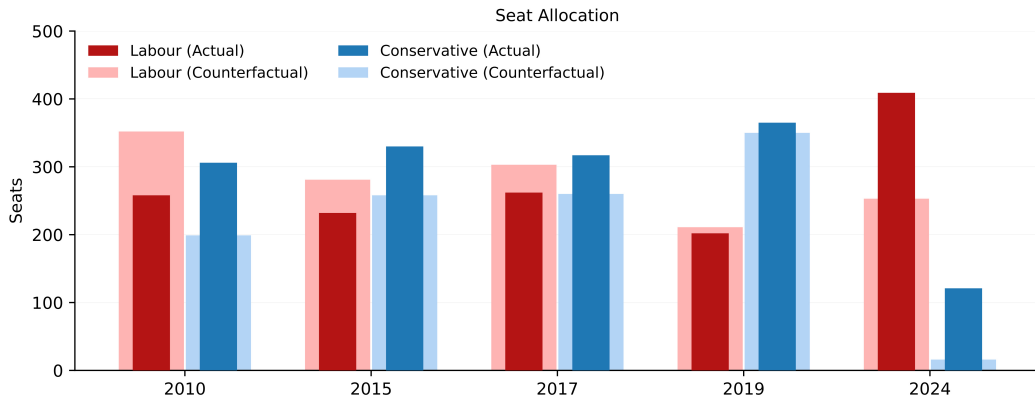
Source: Author's calculations from official election returns.

Long-run pattern: Labour vs Conservative



Source: Author's calculations from official election returns.

Long-run pattern: Labour vs Conservative

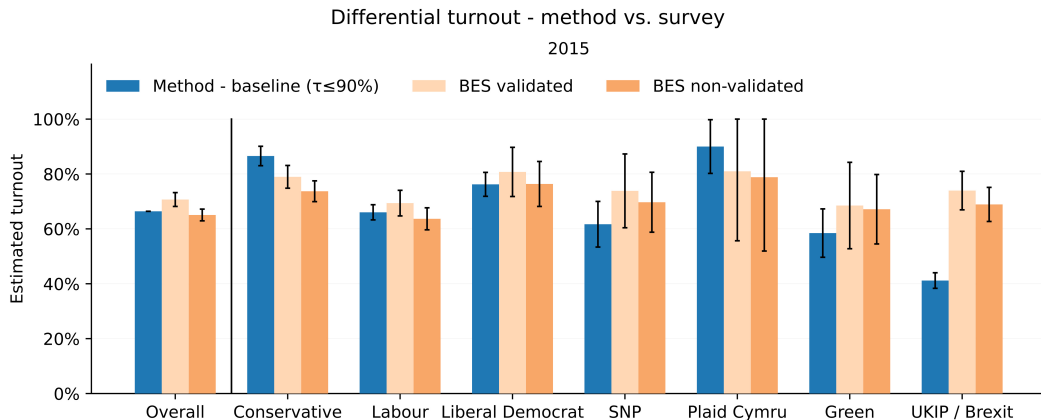


Source: Author's calculations from official election returns.

Validation and robustness

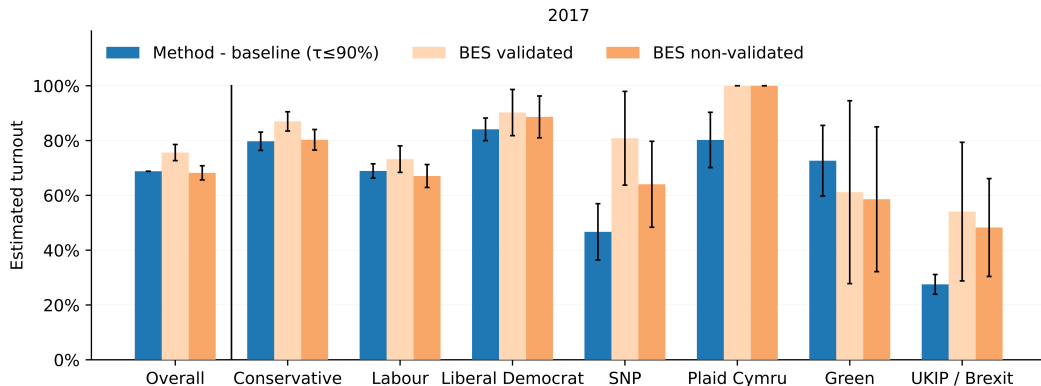
- ▶ Compare administrative-data estimates with British Election Study (BES) turnout measures:
 - ▶ **BES validated**: respondents whose turnout could be verified against electoral registers
 - ▶ **BES non-validated**: full sample with survey weights
- ▶ Sensitivity: $\tau_{\max} \in \{0.85, 0.90, 0.95\}$
- ▶ Compare seat counterfactuals across regional, UK-wide, and survey-based estimates
- ▶ Treat smaller parties with extra caution – sparse support gives less information regionally

Differential turnout: method vs. BES in 2015



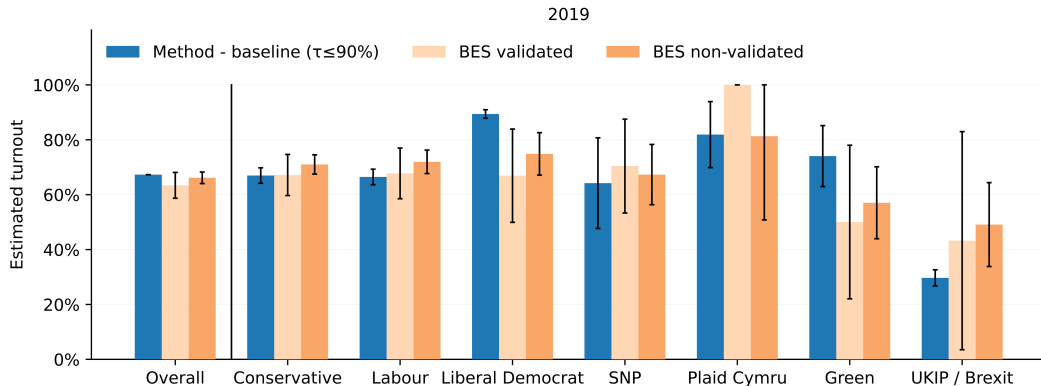
Source: Author's calculations and British Election Study turnout measures.

Differential turnout: method vs. BES in 2017



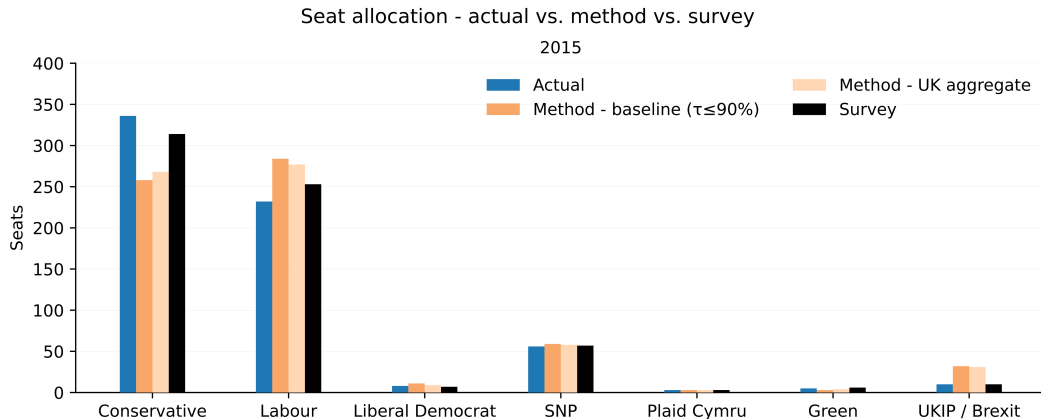
Source: Author's calculations and British Election Study turnout measures.

Differential turnout: method vs. BES in 2019



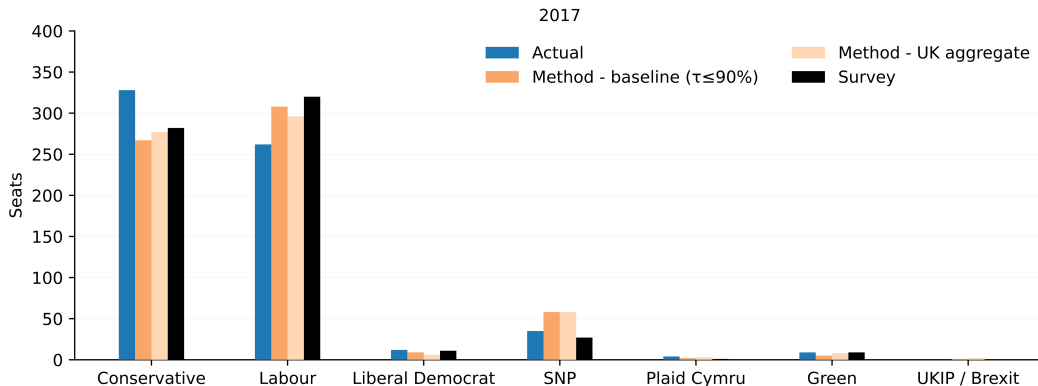
Source: Author's calculations and British Election Study turnout measures.

Seat counterfactuals: method vs. survey in 2015



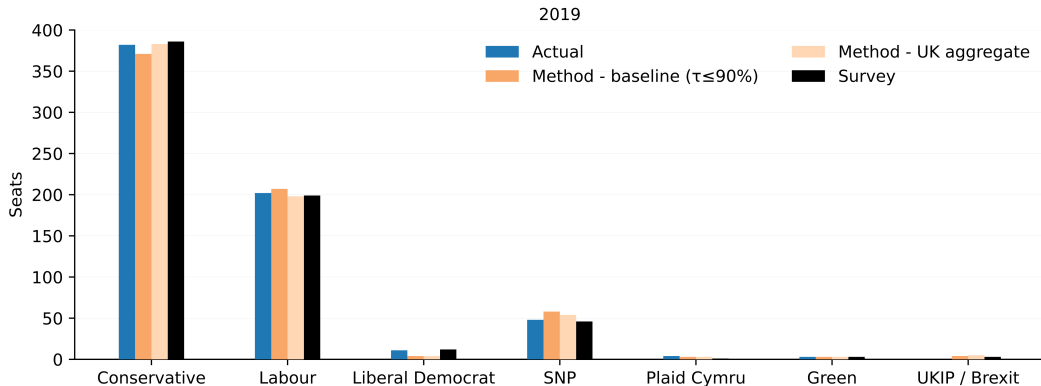
Source: Author's calculations and British Election Study turnout measures.

Seat counterfactuals: method vs. survey in 2017



Source: Author's calculations and British Election Study turnout measures.

Seat counterfactuals: method vs. survey in 2019



Source: Author's calculations and British Election Study turnout measures.

What the results imply

- ▶ Mobilization gaps can determine not just margins, but which party governs
- ▶ FPTP amplifies turnout advantages in marginal constituencies
- ▶ Seat shares may overstate persuasion or realignment when asymmetric abstention is large
- ▶ For campaigns: high returns to registration and GOTV among low-propensity supporters
- ▶ Connects to debates on compulsory voting, voter ID, and registration reform

Caveats

- ▶ Regional homogeneity is a maintained identification assumption
- ▶ Tactical voting and fluid attachments complicate the “latent supporter” construct
- ▶ Counterfactuals abstract from campaign responses and equilibrium effects
- ▶ Uncertainty larger for niche parties and parties with sparse regional support

Broader applicability

- ▶ The framework requires only **votes and electorates**
- ▶ Updates rapidly after each new election
- ▶ Applicable to subnational elections, historical elections, and other countries
- ▶ Cross-national application increasingly feasible (e.g., CLEA)
- ▶ Comparative agenda: how do electoral rules mediate turnout gaps and representation bias

Conclusion

- ▶ Party-specific turnout can be recovered from official returns using simple accounting constraints
- ▶ Britain shows **persistent partisan mobilization gaps** from 2010 to 2024
- ▶ Conservative supporters vote at higher rates than Labour and UKIP/Brexit/Reform UK supporters
- ▶ Universal-turnout counterfactuals would have reversed outcomes in **three of five elections**